Ellipsoidal techniques for reachability analysis: Internal approximation*

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Abstract

For the reach tube of a linear time-varying system with ellipsoidal bounds on the control variable consider the following approximation problem. Find a tight ellipsoid-valued tube *inside* the reach tube that touches it along any prespecified smooth curve on the boundary? We construct this ellipsoidal tube, and show that if the given curve is itself a system trajectory, the tube can be calculated recursively, with minimum computational burden.

Introduction

Recent progress in on-line automation and the study of hybrid systems has renewed interest in the effective computation of reach sets [6, 11, 10, 7, 12, 15, 4]. Methods for reachability analysis of linear systems include those based on ellipsoidal representations [14, 1, 2, 8]. Most papers on this subject deal with *external* ellipsoidal approximations of reach sets. But requirements of guaranteed performance demand the calculation of *internal* approximations. This is a more difficult problem. As indicated in [8], an exact representation of reach sets and reach tubes is possible through the union of a family of ellipsoids. One particular approximation, the volume-optimal internal ellipsoid, was studied in [2]. Other internal ellipsoidal approximations were suggested in [1]. An important open problem is to compute a family of tight internal approximations of reach tubes through such ellipsoid-valued functions.

The present paper solves this problem for time-varying linear systems with ellipsoidal bounds on the control variable. We study the following question: given a reach tube and a smooth curve on its boundary, can you find an ellipsoid-valued tube *inside* the reach tube that touches its boundary precisely along the prespecified curve?

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The solution is unique and we show how to construct it. The computation is burdensome because "fresh" recalculations are required at each instant of time. However, there exists a family of "good" curves for which the solution is recursive. A "good" curve turns out to be any system trajectory. By covering the surface of the reach tube with these trajectories, we can produce the whole reach tube through such internal ellipsoidal tubes, while minimizing the computational burden. The paper relies on relations for internal ellipsoids different from those in [1, 2, 8].

1 Preliminaries.The Reachability Problem.

We consider the linear control system

$$\dot{x} = A(t)x + B(t)u, \quad t_0 \le t \le t_1, \tag{1}$$

where $x \in \mathbb{R}^n$ is the state and $u \in \mathbb{R}^m$ is the control. The matrices $A(t), B(t)$ are continuous and the *controllability property* (see [10]) holds, i.e., the quadratic form

$$W(\tau, t_0) = \int_{t_0}^{\tau} (l, X(\tau, s)B(s)B'(s)X'(\tau, s)l)ds, \; l \in \mathbb{R}^n,$$

is positive definite for any $\tau \in (t_0, t_1]$. Here $X(t, s)$ is the transition matrix,

$$\partial X(t, s)/\partial t = A(t)X(t, s), \;\; X(s, s) = I,$$

and I is the unit matrix. If $A(t) = const, X(t, s) = \exp(A(t - s))$.

The control $u = u(t)$ is any measurable function restricted by the hard bound

$$u(t) \in \mathcal{P}(t), \tag{2}$$

for almost all $t \ge t_0$. The set-valued bound $\mathcal{P}(t)$ is a nondegenerate ellipsoid, continuous in t : $\mathcal{P}(t) = \mathcal{E}(q(t), Q(t))$, where

$$\mathcal{E}(q(t), Q(t)) = \{u : (u - q(t), Q^{-1}(t)(u - q(t))) \le 1\}, \tag{3}$$

with function $q(t) \in \mathbb{R}^m$ (the center of the ellipsoid) and positive definite matrix function $Q(t) \in \mathbb{R}^{m\times m}$ (the matrix of the ellipsoid) continuous in t. Its *support function* ([13]) is

$$\rho(l|\mathcal{E}(q(t), Q(t))) = \max\{(l, x)|x \in \mathcal{E}(q(t), Q(t))\} = (l, q(t)) \; + \; (l, Q(t)l)^{1/2}.$$

Definiton 1.1 *Given position* $\{t_0, x_0\}$, *the* **reach set** *("attainability domain")* $\mathcal{X}(\tau, t_0, x^0)$ at time $\tau > t_0$ *from this position is the set*

$$\mathcal{X}[\tau] = \mathcal{X}(\tau, t_0, x^0) = \cup\{x(\tau, t_0, x^0|u(\cdot))|u(\cdot) \in \mathcal{P}(\cdot)\}$$

of all states $x[\tau]$ *reachable at time* τ, *with* $x(t_0) = x^0$, *over all the possible controls* u *that satisfy (2). The set–valued function* $\mathcal{X}[\tau] = \mathcal{X}(\tau, t_0, x^0)$ *is the* **reach tube** *from* x^0. *The reach set* $\mathcal{X}(\tau, t_0, \mathcal{X}^0)$ *at time* τ *from set* $X^0 = \mathcal{X}(t_0)$ *is the set*

$$\mathcal{X}(\tau, t_0, X^0) = \cup\{\mathcal{X}(\tau, t_0|x^0 \in \mathcal{X}^0\}.$$

The set–valued function $\mathcal{X}[\tau] = \mathcal{X}(\tau, t_0, \mathcal{X}_0)$ *is the reach tube from set* $\mathcal{X}^0$.

The following properties may be checked directly.

Lemma 1.1 *The set-valued function $\mathcal{X}(t, t_0, \mathcal{X}^0)$ satisfies the semigroup property*

$$X(t, t_0, X^0) = X(t, \tau, X(\tau, t_0, X^0)). \tag{4}$$

In the sequel it is assumed that $\mathcal{X}^0 = \mathcal{E}(x^0, X^0)$ is an ellipsoid.

The reach set $\mathcal{X}(t, t_0, \mathcal{E}(x^0, X^0))$ is also given as a set-valued ("Aumann") integral.

Lemma 1.2 *The following relation holds:*

$$\mathcal{X}(t, t_0, \mathcal{E}(x^0, X^0)) = x^\star(t) + X(t, t_0)\mathcal{E}(0, X^0) + \int_{t_0}^{t} X(t, s)\mathcal{E}(0, B(s)Q(s)B'(s))ds, \tag{5}$$

where

$$x^\star(t) = X(t, t_0)x^0 + \int_{t_0}^{t} X(t, s)B(s)q(s)ds.$$

A standard calculation of convex analysis yields the following result (see, for example [6, 8]).

Lemma 1.3 *The support function*

$$\rho(l|\mathcal{X}(t, t_0, \mathcal{E}(x^0, X^0)) = (l, x^\star(t)) + (l, X(t, t_0)X^0X'(t, t_0)l)^{1/2} + \tag{6}$$

$$+ \int_{t_0}^{t} (l, X(t, s)B(s)Q(s)B'(s)X'(t, s)l)^{1/2}ds.$$

The next lemma now follows.

Lemma 1.4 *$\mathcal{X}[t] = \mathcal{X}(t, t_0, \mathcal{E}(x^0, X^0))$ is a convex compact set in $\mathbb{R}^n$, continuous in t.*

The continuity of $\mathcal{X}[t]$ means that its support function $\rho(l|\mathcal{X}[t])$ is continuous in t uniformly in l with $(l, l) \leq 1$.

Recall an important property of any point x^* on the boundary $\partial\mathcal{X}[\tau]$ of the reachability set $\mathcal{X}[\tau] = X(\tau, t_0, \mathcal{E}(x^0, X^0))$.[1] Corresponding to x^* there is a *support vector* l^* such that

$$(l^*, x^*) = \rho(l^*|\mathcal{X}[\tau]), \tag{7}$$

The initial state $x(t_0) = x^{*0} \in \mathcal{E}(x^0, X^0)$ and the control $u^*(t)$ which steers the system from state $x(t_0) = x^{*0}$ to $x(\tau) = x^*$ are specified in the following fact (see [6, 10]).

[1] The boundary of $\mathcal{X}[\tau]$ is the $\partial\mathcal{X}[\tau] = \mathcal{X}[\tau] \setminus int\mathcal{X}[\tau]$. Under the controllability assumption $\mathcal{X}[\tau]$ always has a nonvoid interior $int\mathcal{X}[\tau] \neq \emptyset$.

Theorem 1.1 *Suppose $x(\tau) = x^* \in \partial\mathcal{X}[\tau]$. Then the control $u = u^*(t)$ and the initial state $x(t_0) = x^{*0}$ that yield a unique trajectory $x^*[t]$ which reaches $x^* = x^*[\tau]$ from $x^{*0} = x(t_0)$ satisfy the "maximum principle" for almost all $t \in [t_0, \tau]$:*

$$(l^*, X(\tau,t)B(t)u^*(t)) = \max\{(l^*, X(\tau,t)B(t)u) | u \in \mathcal{E}(q(t), Q(t))\} = \tag{8}$$
$$= (l^*, q(t)) + (l^*, X(\tau,t)B(t)Q(t)B'(t)X'(\tau,t)l^*)^{1/2},$$

and the extremal relation

$$(l^*, x^{*0}) = \max\{(l^*, x) | x \in \mathcal{E}(x^0, X^0)\} = (l^*, x^0) + (l^*, X^0 l^*)^{1/2}, \tag{9}$$

where l^* is the support vector for set $\mathcal{X}[t]$ at x^* that satisfies relation (7).

Exact calculation of reach sets is difficult. Among the approximation methods for these problems are those that rely on *ellipsoidal techniques*. Although the initial set $\mathcal{E}(x^0, t_0)$ and the control set are ellipsoids, the reach set $\mathcal{X}[t] = \mathcal{X}(t, t_0, \mathcal{E}(x^0, X^0))$ will not generally be an ellipsoid. As discussed in [8], it may be approximated both externally and internally by ellipsoids $\mathcal{E}_+$ and $\mathcal{E}_-$ with $\mathcal{E}_- \subseteq \mathcal{X}[t] \subseteq \mathcal{E}_+$.

These approximations are *tight* if for any ellipsoid $\mathcal{E}$ the inclusion $\mathcal{X}[t] \subseteq \mathcal{E} \subseteq \mathcal{E}_+$ implies $\mathcal{E} = \mathcal{E}_+$, and $\mathcal{E}_- \subseteq \mathcal{E} \subseteq \mathcal{X}[t]$ implies $\mathcal{E} = \mathcal{E}_-$. The properties of tight *external* approximations were studied in [9]. We obtain tight *internal* approximations in this paper.

The idea of exactly describing the reach set $\mathcal{X}[t]$ through the union of a parametrized family of internal ellipsoids was discussed in [8]. Our aim will be to specify the tight internal ellipsoids that touch the reach set $\mathcal{X}[t]$ from inside along specified curves on its boundary $\partial\mathcal{X}[t]$.

2 The internal ellipsoids

To approximate the sum of two ellipsoids $\mathcal{E}(0, Q_1)$ and $\mathcal{E}(0, Q_2)$, take the ellipsoid $\mathcal{E}(0, Q_-[S])$, where

$$Q_-[S] = (S_1 Q_1^{1/2} + S_2 Q_2^{1/2})'(S_1 Q_1^{1/2} + S_2 Q_2^{1/2})$$

and S_1, S_2 are orthogonal matrices [3], with $S_1' S_1 = I = S_2' S_2$. Then

$$(\rho(l|\mathcal{E}(0, Q_-[S])))^2 = (l, Q_-[S]l) = (l, Q_1 l) + (l, Q_2 l) + 2(S_1 Q_1^{1/2} l, S_2 Q_2^{1/2} l)$$
$$\leq (l, Q_1 l) + (l, Q_2 l) + 2(l, Q_1 l)^{1/2}(l, Q_2 l)^{1/2} = ((l, Q_1 l)^{1/2} + (l, Q_2 l)^{1/2})^2$$

for any vector $l \in \mathbb{R}^n$ with equality iff

$$S_1 Q_1^{1/2} l = k S_2 Q_2^{1/2} l$$

for some value k. The last assertion follows from the Hölder inequality. For the sum of m ellipsoids $\mathcal{E}(0, Q_i), i = 1, ..., m$ we have the following proposition.

Theorem 2.1 *We have*

$$\mathcal{E}_-(0, Q[S(m)]) \subseteq \sum_{i=1}^{m} \mathcal{E}_i(0, Q_i),$$

where

$$Q[S(m)] = (\sum_{i=1}^{n} S_i Q^{1/2})'(\sum_{i=1}^{m} S_i Q^{1/2}),$$

$S[(m)] = \{S_1, \ldots, S_m\}$ *are any orthogonal matrices, $S_i' S_i = I$.*

Indeed, using the Hölder inequality,

$$(\rho(l|\mathcal{E}(0,Q_-[S(m)])))^2 = (l,Q_-[S(m)]l) = \sum_{i=1}^m (l,Q_i l) + \sum_{i\neq j}^m (S_i Q_i^{1/2} l, S_j Q_j^{1/2} l)$$

$$\leq \sum_{i=1}^m (l,Q_i l) + \sum_{i\neq j} (l,Q_i l)^{1/2}(l,Q_j l)^{1/2} = (\sum_{i=1}^m (l,Q_i l)^{1/2})^2,$$

for any $l \in \mathbb{R}^n$. This proves the theorem.

Here equality is attained for a given vector l iff

$$S_i Q_i^{1/2} l = k_{ij} S_j Q_j^{1/2} l, \;\; k_{ij} = k_{ji}^{-1},$$

for some values k_{ij} (that depend on l). This happens, in its turn, iff there exists a vector $p \neq 0$ and and an array of numbers λ_i such that

$$S_i Q_i^{1/2} l = \lambda_i p, \;\; i = 1,\ldots,m. \tag{10}$$

This proves the next proposition.

Theorem 2.2 *The inequality*

$$(l,Q[S(m)]l) = (l,(\sum_{i=1}^m S_i Q^{1/2})'(\sum_{i=1}^m S_i Q^{1/2})l) \leq \sum_{i=1}^m (l,Q_i l)^{1/2}, \tag{11}$$

is true for any $l \in \mathbb{R}^n$ and any array $S[m]$ of orthogonal matrices $S_i, i = 1,\ldots,m,$. It turns into an equality for a given $l \in \mathbb{R}^n$ iff the matrices S_i are such that (10) is fulfilled for the given l for some vector p and numbers λ_i, $i = 1,\ldots,m$.

The last theorem indicates *the tightness conditions* for the internal ellipsoidal approximation $\mathcal{E}_-(0,Q[S(m)])$ of the sum of ellipsoids $\mathcal{E}(0,Q_i)$.

Remark 2.1 The previous relations were derived for the sum of ellipsoids centered at $q_i = 0$. They also hold for ellipsoids $\mathcal{E}(q_i,Q_i)$, with $q_i \neq 0$, if we substitute $\mathcal{E}_-(0,Q_-[S(m)])$ by $\mathcal{E}_-(q(m),Q_-[S(m)])$, where

$$q(m) = \sum_{i=1}^m q_i. \tag{12}$$

We now look at the internal approximation of the sum

$$\mathcal{E}(0,Q_0) + \int_{t_0}^t \mathcal{E}(0,Q(s))ds,$$

of an ellipsoid $\mathcal{E}(0,Q_0)$ and a set-valued integral of an ellipsoid-valued function $\mathcal{E}(0,Q(s))$.

Taking the Riemanian sum

$$I(\Sigma_N) = \sum_{i=1}^n \mathcal{E}(q(\tau_i),Q(\tau_i))\sigma_i, \tag{13}$$

generated by the partition

$$\Sigma_N = \{\tau_0 = t_0, \tau_1, \ldots, \tau_N = t\}, \quad \sigma_i = \tau_i - \tau_{i-1}, \quad i = 1, \ldots, N,$$

we apply the results of Theorems 2.1, 2.2. Therefore the matrix

$$Q_-^{(N)}(t) = \left(S_0 Q_0^{1/2} + \sum_{i=1}^{N} S_i Q_i^{1/2}(\tau_i)\sigma_i\right)' \left(S_0 Q_0^{1/2} + \sum_{i=1}^{N} S_i Q_i^{1/2}(\tau_i)\sigma_i\right) \tag{14}$$

satisfies the inequality

$$(l, Q_-^{(N)}(t)l) \leq ((l, Q_0 l)^{1/2} + \sum_{i=1}^{N}(l, Q(\tau_i)l)^{1/2}\sigma_i)^2, \tag{15}$$

for any $l \in \mathbb{R}^n$. Equality is reached for a given $l \in \mathbb{R}^n$ iff the orthogonal matrices S_0, S_i satisfy

$$S_i(\tau_i)Q^{1/2}(\tau_i)l = \lambda(\tau_i)S_0 Q_0^{1/2} l, \tag{16}$$

for some values $\lambda(\tau_i), i = 1, \ldots, N$. One may figure out immediately from (16) that these values actually are

$$\lambda_i(\tau) = (l, Q(\tau_i)l)^{1/2}(l, Q_0 l)^{-(1/2)}. \tag{17}$$

Passing to the limit in (13)-(15), with

$$N \to \infty, \quad \sigma[N] = \max\{\sigma_i | i = 1, \ldots, N\} \to 0,$$

we come to the next proposition.

Theorem 2.3 *(i) The following inclusion is true*

$$\mathcal{E}(0, Q_-(t)) \subseteq \mathcal{E}(0, Q_0) + \int_{t_0}^{t} \mathcal{E}(0, Q(\tau))d\tau, \tag{18}$$

for any matrix

$$Q_-(t) = \left(S_0 Q_0^{1/2} + \int_{t_0}^{t} S(\tau)Q(\tau)^{1/2}d\tau\right)' \left(S_0 Q_0^{1/2} + \int_{t_0}^{t} S(\tau)Q(\tau)^{1/2}d\tau\right),$$

where $S_0 S_0' = I, S(\tau)'S(\tau) \equiv I$ are orthogonal matrices that vary continuously in time.

(ii) For a given $l \in \mathbb{R}^n$ relation (15) turns into an equality iff matrices $S_0, S(\tau)$ are chosen such that

$$S(\tau)Q^{1/2}(\tau)l = \lambda(\tau)S_0 Q_0^{1/2} l \tag{19}$$

for all $\tau \in [t_0, t]$ for some scalar function $\lambda(\tau)$.

Theorem 2.3 gives conditions for the internal ellipsoids $\mathcal{E}(0, Q_-(\tau))$ to be *tight*. One may note immediately that function $\lambda(\tau)$ of (19) is actually

$$\lambda(\tau) = (l, Q(\tau)l)^{1/2}(l, Q_0 l)^{-(1/2)}.$$

The proof of the theorem follows by taking the limit in (13) in view of the inequality

$$\left(l, \left(S_0 Q_0^{1/2} + \int_{t_0}^{t} S(\tau)Q^{1/2}(\tau)d\tau\right)' \left(S_0 Q_0^{1/2} + \int_{t_0}^{t} S(\tau)Q^{1/2}(\tau)d\tau\right)l\right) \tag{20}$$

$$\leq \left(\int_{t_0}^{t} (l, Q(\tau) l)^{1/2} d\tau + (l, Q_0 l)^{1/2} \right)^2$$

which follows by taking limits in in (14),(15), with $\sigma[N] \to 0, N \to \infty$.

Remark 2.2. For ellipsoids $\mathcal{E}(q_0, Q_0), \mathcal{E}(q(t), Q(t))$ with nonzero centers $q_0, q(t)$, Theorem 2.3 remains true with substitution of $\mathcal{E}_-(0, Q_-(t))$ by $\mathcal{E}(q_-(t), Q_-(t))$, where

$$q_-(t) = q_0 + \int_{t_0}^{t} q(s) ds. \tag{21}$$

The functions $q_-(t), Q_-(t)$ may be described by differential equations. Differentiating $Q_-(t)$ gives

$$\dot{Q}_-(t) = (S(t) Q^{1/2}(t))' Q_*(t) + Q_*(t) (S(t) Q^{1/2}(t)),$$

where

$$Q_*(t) = S_0 Q_0^{1/2} + \int_{t_0}^{t} S(\tau) Q^{1/2}(\tau) d\tau$$

Introducing the notation

$$H(t) = Q_*^{-1}(t) S(t) Q^{1/2}(t) = Q_*^{-1}(t) \dot{Q}_*(t),$$

we arrive at

$$\dot{Q}_-(t) = H'(t) Q_-(t) + Q_-(t) H(t), \quad Q(t_0) = Q_0. \tag{22}$$

A differentiation of (21) also gives

$$\dot{q}_-(t) = q(t), \quad q_-(t_0) = q_0. \tag{23}$$

Lemma 2.1 *The ellipsoid $\mathcal{E}(q_-(t), Q_-(t))$is described by differential equations (22),(23).*

Relation (22) allows an infinitesimal expansion

$$Q_-(t+\sigma) = Q_-(t) + \sigma(\dot{Q}_*' Q_*(t) + Q_*(t)' \dot{Q}_*) + o(\sigma) \tag{24}$$

where $o(\sigma)/\sigma \to 0$ with $\sigma \to 0$, or equivalently,

$$(l, Q_-(t+\sigma) l) = (l, Q_-(t) l) + \sigma^2 (l, (\dot{Q}_*' Q_*(t) + Q_*(t)' \dot{Q}_*) l) + o(\sigma, l), \tag{25}$$

where $o(\sigma, l)$ stands for a function that satisfies $o(\sigma, l)/\sigma \to 0$ with $\sigma \to 0$, uniformly in $\{l : (l, l) \leq 1\}$.

On the other hand, the relation for $Q_-(t)$ of Theorem 2.3 gives

$$\rho^2(l | \mathcal{E}(0, Q_-(t+\sigma))) = (l, Q_-(t+\sigma) l) = (l, Q_-(t) l) + \tag{26}$$

$$+ \left(l, \left(\int_t^{t+\sigma} S(\tau) Q(\tau)^{1/2} d\tau \right)' Q_*(t) + Q_*'(t) \left(\int_t^{t+\sigma} S(\tau) Q(\tau)^{1/2} d\tau \right) l \right) + o_1(t, l)$$

$$\leq (\rho(l | \mathcal{E}(0, Q_0)) + + \int_{t_0}^{t+\sigma} \rho(l | \mathcal{E}(0, Q(\tau))) d\tau)^2 = \rho^2(l | \mathcal{X}[t]),$$

with equality attained for a given l (up to terms of order >1 in σ) iff there exists $\lambda(t)$ such that

$$\left(\int_t^{t+\sigma} S(\tau)Q(\tau)^{1/2}d\tau\right)l = \lambda(t)Q_*(t)l. \tag{27}$$

As one may observe, here

$$\lambda^2(t) = (l, Q(t)l)/(l, Q_-(t)l) + o_2(t,l),$$

and also

$$\int_t^{t+\sigma} S(\tau)Q(\tau)^{1/2}d\tau = \dot{Q}_*(t) + o_3(t,l).$$

Since equalities (25),(26) ensure, (up to terms of order >1 in σ) that $\mathcal{E}(0, Q_-(t+\sigma))$ touches $\mathcal{X}[t+\sigma]$ at point of support of vector l, we conclude that $\dot{Q}_*(t)l = \lambda(t)Q_*l$ or

$$H(t)l = Q_*^{-1}\dot{Q}_*(t)l = \lambda(t)l, \tag{28}$$

where $\lambda(t) = (l, Q(t)l)^{1/2}/(l, Q_-(t)l)^{1/2}$.

3 The internal ellipsoids for reach sets

We return to (1)-(3), at first with $B(t) \equiv I$. (See Remark 3.2 at the end of this section.) Our goal is to find the internal ellipsoid $\mathcal{E}(q_-(t), Q_-(t))$ for the set

$$\mathcal{X}[t] = X(t,t_0)\mathcal{E}(q_0, Q_0) + \int_{t_0}^t X(t,\tau)\mathcal{E}(q(t), Q(\tau))d\tau.$$

The formula of Theorem 2.3 for the matrix $Q_-(t)$ now is[2]

$$Q_-(t) = \tag{29}$$

$$X(t,t_0)\left(Q_0S_0'(t_0) + \int_{t_0}^t X(t_0,\tau)Q^{1/2}(\tau)S'(\tau)d\tau\right)\left(S_0(t_0)Q_0 + \int_{t_0}^t S(\tau)Q^{1/2}(\tau)X'(t_0,\tau)\right)X'(t,t_0),$$

and

$$q_-(t) = X(t,t_0)q_0 + \int_{t_0}^t X(t,\tau)q(\tau)d\tau. \tag{30}$$

So Theorem 2.3 transforms into the following.

Theorem 3.1 *The internal ellipsoids for the reach set $\mathcal{X}[t]$ satisfy*

$$\mathcal{E}(q_-(t), Q_-(t)) \subseteq \tag{31}$$

$$\mathcal{E}(X(t,t_0)q_0, X(t,t_0)Q_0X'(t,t_0)) + \int_{t_0}^t \mathcal{E}\left(X(t,\tau)q(\tau)d\tau, X(t,\tau)Q(\tau)X'(t,\tau)d\tau\right) = \mathcal{X}[t],$$

where $Q_-(t_0), q_-(t_0)$ are given by (29),(30), with $S_0, S(\tau)$ being any orthogonal matrices and $S(\tau)$ continuous in time.

The tightness conditions transform as follows.

[2]Since for a matrix-valued map X we have $X\mathcal{E}(q,Q) = \mathcal{E}(Xq, XQX')$.

Theorem 3.2 *For a given instant t the internal ellipsoid $\mathcal{E}(q_-(t), Q_-(t))$ is tight and touches $\mathcal{X}[t]$ at the point of support x^* of the tangent hyperplane generated by the given vector l^*, i.e.,*

$$\rho(l^*|\mathcal{X}[t]) = (l^*, q_-(t)) + \int_{t_0}^{t} (l^*, X(t,\tau)Q(\tau)X'(t,\tau)l^*)^{1/2} d\tau = \tag{32}$$

$$\rho(l^*|\mathcal{E}(q_-(t), Q_-(t))) = (l^*, q_-(t)) + (l^*, Q_-(t)l^*)^{1/2} = (l^*, x^*),$$

iff $S_0, S(\tau)$ satisfy the relation

$$S(\tau)Q^{1/2}(\tau)X'(t,\tau)l^* = \lambda(\tau)S_0 Q_0^{1/2} X'(t,t_0)l^*, \;\; t_0 \le \tau \le t, \tag{33}$$

for some function $\lambda(\tau)$.

Direct calculation indicates the following.

Lemma 3.1 *The function $\lambda(\tau)$ of Theorem 3.2 is given by*

$$\lambda(\tau) = (l^*, X(t,\tau)Q(\tau)X'(t,\tau)l^*)^{1/2} / (l^*, X(t,t_0)Q_0X'(t,t_0)l^*)^{1/2}, \; t_0 \le \tau \le t. \tag{34}$$

Theorems 3.1, 3.2 were formulated *for a fixed instant of time t and a fixed support vector l^**. It is important to realize what happens if l^* varies in time.

Problem 3.1. Given a vector function $l^*(t)$, continuously differentiable in t, find an internal ellipsoid $\mathcal{E}(q_-(t), Q_-(t)) \subseteq \mathcal{X}[t]$ that ensures **for all** $t \ge t_0$, the equality

$$\rho(l^*(t)|\mathcal{X}[t]) = \rho(l^*(t)|\mathcal{E}(q_-(t), Q_-(t)) = (l^*(t), x^*(t)), \tag{35}$$

so that the supporting hyperplane for $\mathcal{X}[t]$ generated by $l^*(t)$, namely, the plane $(x - x^*(t), l^*(t)) = 0$ that touches $\mathcal{X}[t]$ at point $x^*(t)$, is also a supporting hyperplane for $\mathcal{E}(q_-(t), Q_-(t))$ and touches it at the same point.

We solve this problem using Theorems 3.1, 3.2. However, the functions $S(\tau), \lambda(\tau)$ used for the parametrization in (29),(33), will now be functions of two variables, τ and t, since (35) must hold for each $t \ge t_0$ (and therefore S_0 should also depend on t). We may therefore still apply Theorems 3.1, 3.2 but now with $S_0, S(\tau), \lambda(\tau)$ replaced by $S_{0t}, S_t(\tau), \lambda_t(\tau)$.

Theorem 3.3 *With $l = l^*(t)$ given, the solution to Problem 3.1 is an ellipsoid $\mathcal{E}(q_-(t), Q_-(t))$, where*

$$Q_-(t) = \tag{36}$$

$$= X(t,t_0)\left(Q_0^{1/2}S'_{0t}(t_0) + \int_{t_0}^{t} X(t_0,\tau)Q^{1/2}(\tau)S'_t(\tau)d\tau\right)\left(S_{0t}(t_0)Q_0^{1/2} + \int_{t_0}^{t} S_t(\tau)Q^{1/2}(\tau)X'(t_0,\tau)\right)X'(t,t_0).$$

with $S_0, S_t(\tau)$ satisfying

$$S_t(\tau)Q^{1/2}(\tau)X'(t,\tau)l^*(t) = \lambda_t(\tau)S_{0t}Q_0^{1/2}X'(t,t_0)l^*(t), \tag{37}$$

and $S'_{0t}S_{0t} = I$; $S'_t(\tau)S_t(\tau) \equiv I$ for all $t \ge t_0$, $\tau \in [t_0, t]$, where

$$\lambda_t(\tau) = (l^*(t), X(t,\tau)Q(\tau)X'(t,\tau)l^*(t))^{1/2}(l^*(t), X(t,t_0)Q_0X'(t,t_0)l^*(t))^{1/2}. \tag{38}$$

The proof follows by direct substitution in Theorems 3.1, 3.2. The relations (37), (38) are in a "static" form: the parameters $S_{0t}, S_t(\tau), \lambda_t(\tau), \tau \in [t_0, t]$ must be recalculated "afresh" for each t. We now investigate whether the calculations can be made recursive, without this recalculation.

Remark 3.1. In all the ellipsoidal approximations considered in this paper the center of the approximating ellipsoid is always given by $q_-(t)$ of (30). The discussions shall therefore actually concern only the relations for $Q_-(t)$.

Remark 3.2. If $B(t) \neq I$ we must replace $Q(t)$ by $B(t)Q(t)B'(t)$, $Q^{1/2}(t)$ for $B(t)Q^{1/2}(t)$ and $q(t)$ by $B(t)q(t)$ in all the relations of this and later sections, starting the calculation process from time $t \geq t_0 + \sigma,\ \sigma > 0$ rather then from t_0. The controllability assumption of section 1 ensures that $int\mathcal{X}[t] \neq \emptyset$ and the ellipsoids $\mathcal{E}(q(t), Q_-(t))$ are nondegenerate.

4 Recurrence Relations

We start with a particular function $l^*(t)$.

Assumption 4.1 *The function $l^*(t)$ is of the the following form*

$$l^*(t) = X'(t_0, t)l,$$

with vector $l \in \mathbb{R}^n$ given. If the system is time-invariant, $l^*(t) = e^{-A'(t-t_0)}l$.

In this case $l^*(t)$ is the solution to the adjoint of (1),

$$\dot{l}^* = -A'(t)l^*,\ \ l^*(t_0) = l.$$

Substituting $l^*(t)$ in (37),(38), the relations for calculating $S_t(\tau), \lambda_t(\tau)$ become

$$S_t(\tau)Q^{1/2}(\tau)X'(t_0, \tau)l = \lambda_t(\tau)S_{0t}Q_0^{1/2}l, \quad S_0'S_0 = I,\ S'(\tau)S(\tau) \equiv I \tag{39}$$

and

$$\lambda_t(\tau) = (l, X(t_0, \tau)Q(\tau)X'(t_0, \tau)l)^{1/2}/(l, Q_0 l)^{1/2}. \tag{40}$$

Here the known functions used for calculating $S_t(\tau), \lambda_t(\tau)$ *do not depend on* t, and so the unknown functions $S_t(\tau), \lambda_t(\tau)$ *do not depend on* t either, no matter what is the interval $[t_0, t]$. Therefore, the subscript t in S_{0t}, S_t, λ_t may be dropped.

Differentiating (36) in view of the last remark, we come to

$$\dot{Q}_- = A(t)Q_- + Q_-A'(t) + \dot{Q}_*'Q_* + Q_*'\dot{Q}_*, \tag{41}$$

where

$$Q_*(t) = S_0Q_0^{1/2}X'(t, t_0) + \int_{t_0}^{t} S(\tau)Q^{1/2}(\tau)X'(t, \tau)d\tau,$$

$$\dot{Q}_*(t) = S(t)Q^{1/2}(t), Q_*(t_0) = S_0Q_0.$$

Using the notation

$$H(t) = Q_*^{-1}(t)S(t)Q^{1/2}(t) = Q_*^{-1}(t)\dot{Q}_*(t), \tag{42}$$

we get

$$\dot{Q}_- = A(t)Q_- + Q_-A'(t) + H'(t)Q_-(t) + Q_-(t)H(t), \;\; Q(t_0) = Q_0. \tag{43}$$

The differentiation of (30) also gives

$$\dot{q_-} = A(t)q_- + q(t), \;\; q(t_0) = q_0. \tag{44}$$

This leads to the following theorem.

Theorem 4.1 *Under Assumption 4.1 the solution to Problem 3.1 is given by the ellipsoid* $\mathcal{E}(q_-(t), Q_-(t))$, *where* $Q_-(t), q_-(t)$ *are given by (43),(44), and the functions* $S(t), \lambda(t)$ *involved in the calculation of* $H(t)$ *satisfy together with* S_0 *the relations (39),(40), where the subscript* t *in* S_{0t}, S_t, λ_t *are to be dropped.*

Lemma 4.1 *The function* $H(t) = Q_*^{-1}(t)S(t)Q^{1/2}(t)$ *in (42) may be also expressed with the aid of equation*

$$\dot{Q}_* = Q_*A'(t) + S(t)Q^{1/2}(t), \; Q_*(t_0) = S_0Q_0^{1/2}. \tag{45}$$

This gives the next result.

Lemma 4.2 *The ellipsoid* $\mathcal{E}(q_-(t), Q_-(t))$ *of Theorem 4.1 given by equations (44), (43), (42), (45), depends on the selection of the orthogonal matrix function* $S(t)$ *and for any such* $S(t)$ *the inclusion*

$$\mathcal{E}(q_-(t), Q_-(t)) \subseteq \mathcal{X}[t], \;\; t \geq t_0,$$

is true with equality (35) attained under conditions (39),(40).

Suppose $l(t)$ of Problem 3.1 is the vector function that generates *any* continuous curve of related *support vectors* on the surface of $\mathcal{X}[t]$. Then one has to use formula (36), keeping in mind that $S_{0t}, S_t(\tau)$ depend on t. Upon differentiating (36), (43) transforms into

$$\dot{Q}_- = A(t)Q_- + Q_-A'(t) + H'(t)Q_-(t) + Q_-(t)H(t) + \Phi(t,\cdot), \;\; Q(t_0) = Q_0. \tag{46}$$

where

$$\Phi(t,\cdot) = X(t,t_0)\Big(Q_0^{1/2}(\partial S'_{0t}(t_0)/\partial t) + \int_{t_0}^{t} X(t_0,\tau)Q^{1/2}(\tau)(\partial S'_t(\tau)/\partial t)d\tau\Big)' \times$$

$$\Big((\partial S_{0t}(t_0)/\partial t)Q_0^{1/2} + \int_{t_0}^{t} \partial(S_t(\tau)/\partial t)Q^{1/2}(\tau)X'(t_0,\tau)d\tau\Big)X'(t,t_0).$$

Lemma 4.3 *Under Assumption 4.1 the functional* $\Phi(t,\cdot) \equiv 0$.

We recall from [9] that $l(t)$ generates a curve $x^*(t)$ of support vectors[3] that form a *trajectory* of (1) (with a certain control $u(t)$) iff $l(t)$ is a solution of the equation

$$\dot{l} + A(t)l = k(t)l(t), \tag{47}$$

for some continuous scalar function $k(t)$. This leads to our final result.

[3]The support vectors are those that satisfy condition $(l(t), x^*(t)) = \rho(l(t)|\mathcal{X}[t]), t \geq t_0$.

Theorem 4.2 *Let $l(t)$ generate a curve $x^*(t)$ of related support vectors for the sets $\mathcal{X}[t]$ that form a system trajectory of (1) due to some control $u(t)$. Then Assumption 4.1 is satisfied ($l(t)$ is a "good" curve) and the functional $\Phi(t,\cdot) \equiv 0$.*

A similar property holds for external ellipsoids [9]. Theorem 4.2 and its counterpart in [9] together provide tight internal and external ellisoid tubes that touch any given trajectory on the boundary of the reach tube.

5 Example

Consider the system

$$\dot{x}_1 = x_2, \;\; \dot{x}_2 = u, \tag{48}$$

with $x_1(0) = x_1^0, \; x_2^0(0) = x_2^0; \;\; |u| \leq \mu, \;\; \mu > 0$. Then

$$x_1(t) = x_1^0 + x_2^0 t + \int_0^t (t-\tau)u(\tau)d\tau, \quad x_2(t) = x_2^0 + \int_0^t u(\tau)d\tau.$$

Assume $X^0 = \mathcal{B}_\epsilon(0) = \{x : (x,x) \leq \epsilon^2\}$. The support function

$$\rho(l|\mathcal{X}[t]) = \max\{(l, x(t))| \; |u| \leq \mu, \; x^0 \in X^0\}$$

of the reach set $\mathcal{X}[t] = \mathcal{X}(t, 0, X^0)$ may be calculated directly and is given by

$$\rho(l|\mathcal{X}[t]) = \epsilon(l_1^2 + (l_1 t + l_2)^2)^{1/2} + \int_0^t |l_1(t-\tau) + l_2| d\tau.$$

The boundary $\partial\mathcal{X}[t]$ of the reach set $\mathcal{X}[t]$ is the set of vectors (x_1, x_2) such that

$$\min_l \{\rho(l|\mathcal{X}[t]) - l_1 x_1 - l_2 x_2 | (l,l) = 1\} = 0. \tag{49}$$

The solution to the optimization problem (49) for $x_1 = x_1(t), x_2 = x_2(t)$ may be done directly, similarly to [9]. It is given by minimizers $l_1 = l_1^0(t), l_2 = l_2^0(t)$. Solving (49) for all t gives a parametric representation of the boundary $\partial\mathcal{X}[t]$ of the reach tube $\mathcal{X}[t]$. Namely,

$$x_1(t) = \epsilon l_1(t)/(l_1^2(t) + (l_1(t)t + l_2(t))^2)^{1/2} \;\pm\; \mu(t^2/2 - (tl_1(t) - l_2(t))^2/l_1^2(t)), \tag{50}$$

$$x_2(t) = \epsilon(l_1(t)t + l_2(t))/(l_1(t)^2 + (l_1(t)t + l_2(t))^2)^{1/2} \pm 2\mu(l_2 - tl_1(t))/l_1(t) \pm \mu t,$$

where $l \in \mathbb{R}^2, t \geq t_0$. Here for each t, $l^0 = l(t)$ is the support vector to $\mathcal{X}[t]$ at point $x(t) \in \partial\mathcal{X}[t]$. Moreover, with t fixed, and $x = x(t)$ running along $\partial\mathcal{X}[t]$ (which is a closed curve in $\mathbb{R}^2$ for a given t), its support vector $l^0 = l(t)$ will sweep out *all* directions in $\mathbb{R}^2$. Therefore, considering any function $l(t)$ of t with $t \geq t_0$, we may be sure that there exists a corresponding trajectory $x(t) \in \partial\mathcal{X}[t]$ for $t \geq t_0$.

Proceeding further, we select $l(t)$ satisfying Assumption 4.1, namely, as $l(t) = e^{-A't}l^*$. This gives $l_1(t) = l_1^*, \; l_2(t) = l_2^* - tl_1^*$, and (5) simplifies to

$$x_1(t) = \epsilon l_1^*/(l_1^{*2} + l_2^{*2})^{1/2} \;\pm\; \mu(t^2/2 - (tl_1^* - l_2^{*2}(t))^2/l_1^{*2}(t)), \tag{51}$$

$$x_2(t) = \epsilon l_2^*/(l_1^{*2} + l_2^{*2})^{1/2} \pm 2\mu(l_2^* - tl_1^*)/l_1^* \pm \mu t.$$

These relations depend only on the two-dimensional vector l^*. They produce a parametric family of curves $\{x_1(t), x_2(t)\}$ that cover all the surface of the reach tube $\mathcal{X}[t]$ so that vectors $x(t) = \{x_1(t), x_2(t)\}$ are the points of support for the hyperplanes generated by vectors $l(t) = \{l_1^*, -tl_1^* + l_2^*\}$. The reach tube that starts at $X^0 \neq \{0\}$ with these curves on its surface is shown in fig.1.

We now construct the tight internal ellipsoidal approximations for $\mathcal{X}[t]$ that touch the boundary $\partial\mathcal{X}[t]$ *from inside* at points of support taken for a given vector $l = l^*$.

The support function $\rho(\mathcal{X}[t])$ may be rewritten as

$$\rho(\mathcal{X}[t]) = \epsilon(l^*, Q(t)l^*)^{1/2} + \mu \int_0^t (l^*, Q(\tau)l^*)^{1/2} d\tau, \tag{52}$$

where

$$Q(\tau) = \begin{pmatrix} \tau^2, & -\tau \\ -\tau, & 1 \end{pmatrix}$$

and $Q^{1/2}(\tau) = Q(\tau)(1+\tau^2)^{-1/2}$.

According to (36) and in view of Assumption 4.1, we have (taking $S_0 = I$),

$$Q_-(\tau) = \left(\epsilon I + \int_0^t Q^{1/2}(\tau) S'(\tau) d\tau\right)' \left(\epsilon I + \int_0^t S(\tau) Q^{1/2}(\tau) d\tau\right), \tag{53}$$

where the matrix $S(\tau)$ must satisfy the conditions

$$S'(\tau)S(\tau) = I, \;\; S(\tau)Q^{1/2}(\tau)l^* = \epsilon\lambda(\tau)l^*, \;\; \tau \geq 0. \tag{54}$$

for some $\lambda(\tau)$. Calculations give

$$\epsilon\lambda^2(\tau) = (l^*, Q(\tau)l^*)(l^*, l^*)^{-1} = (l_1^*\tau - l_2^*)^2(l_1^{*2} + l_2^{*2})^{-1}. \tag{55}$$

Denote

$$p(\tau) = Q^{1/2}(\tau)l^* = \begin{pmatrix} \tau^2 l_1^* - \tau l_2^* \\ -\tau l_1^* + l_2^* \end{pmatrix} (1+\tau^2)^{-1/2} = r_p(1+\tau^2)^{-1/2} \begin{pmatrix} \cos\phi_p(\tau) \\ \sin\phi_p(\tau) \end{pmatrix},$$

where

$$r_p(\tau) = |l_1^*\tau - l_2^*|(1+\tau^2)^{1/2}, \;\; \phi_p(\tau) = \pm\arccos(\tau^2 l_1^* - \tau l_2^*)/r_p = \arccos(\tau/(1+\tau^2)^{1/2})$$

and also

$$l^* = (l^*, l^*)^{1/2} \begin{pmatrix} \cos\phi_l(\tau) \\ \sin\phi_l(\tau) \end{pmatrix}, \;\; \phi_l = \pm\arccos(l_1^*/(l_1^{*2} + l_2^{*2})^{1/2}).$$

Selecting further the orthogonal matrix-valued function $S(\tau)$ as

$$S(\tau) = \begin{pmatrix} \cos\alpha(\tau), -\sin\alpha(\tau) \\ \sin\alpha(\tau), \cos\alpha(\tau) \end{pmatrix},$$

we may rewrite the second relation (54) as

$$\begin{pmatrix} \cos(\phi_p(\tau) + \alpha(\tau)) \\ \sin(\phi_p(\tau) + \alpha(\tau)) \end{pmatrix} r_p(\tau)(1+\tau^2)^{1/2} = \epsilon\lambda(\tau)(l_1^{*2} + l_2^{*2})^{1/2} \begin{pmatrix} \cos\phi_l(\tau) \\ \sin\phi_l(\tau) \end{pmatrix}, \tag{56}$$

where $\tau \in [t_0, t]$. Here $\alpha(\tau)$ has to be selected from the equality

$$\phi_p(\tau) + \alpha(\tau) = \phi_l(\tau), \quad \tau \in [t_0, t], \tag{57}$$

and $\lambda(\tau)$ is given by (55). Equations (56), (57) need no recalculation for new values of t.

Thus we have found an orthogonal matrix function

$$S(\tau) = \begin{pmatrix} \cos(\phi_l(\tau) - \phi_p(\tau)), -\sin(\phi_l(\tau) - \phi_p(\tau)) \\ \sin(\phi_l(\tau) - \phi_p(\tau)), \cos(\phi_l(\tau) - \phi_p(\tau)) \end{pmatrix}, \tag{58}$$

that depends on l^*, is continuous in τ and satisfies (56).

Matrix $Q_-(t)$ may now be calculated from the equations

$$\dot{Q}_- = \dot{Q}'_*(t)Q_*(t) + Q_*(t)\dot{Q}_*(t), \quad Q(0) = \epsilon^2 I, \tag{59}$$

where

$$\dot{Q}_* = S'(t)Q^{1/2}(t), \quad Q_*(0) = \epsilon I.$$

The internal ellipsoids for the reach set $\mathcal{X}[t] = \mathcal{X}(t, 0, X^0)$ are shown in Figures 2-4 for $X^0 = \mathcal{E}(0, \epsilon I)$ with ϵ increasing from $\epsilon = 0$ (Figure 2) to $\epsilon = 0.175$ (Figure 3) to $\epsilon = 1$ (Figure 4). The tube in Figure 1 corresponds to the ϵ of Figure 3. One may also observe the exact reach sets, $\mathcal{X}(t, 0, 0)$ taken for $\epsilon = 0$, inside the sets $\mathcal{X}(t, 0, X^0) = \mathcal{X}[t]$ in Figures 3,4.

6 Conclusion

The calculation of reach sets for linear control systems is needed to determine guaranteed performance bounds and in the study of hybrid systems. This paper shows how to calculate tight internal ellipsoidal approximations, paralleling the calculation of external ellipsoidal approximations in [9]. Through any smooth curve on the boundary of the reach set there passes a unique tight internal ellipsoidal approximation, but its calculation is generally expensive. However, if the smooth curve is itself a trajectory, the calculation is recursive, just as in the case of external approximations. Since such trajectories cover the boundary of the reach set, we now have the means to effectively compute internal and external approximations to the reach set with arbitrary accuracy.

Figure 1

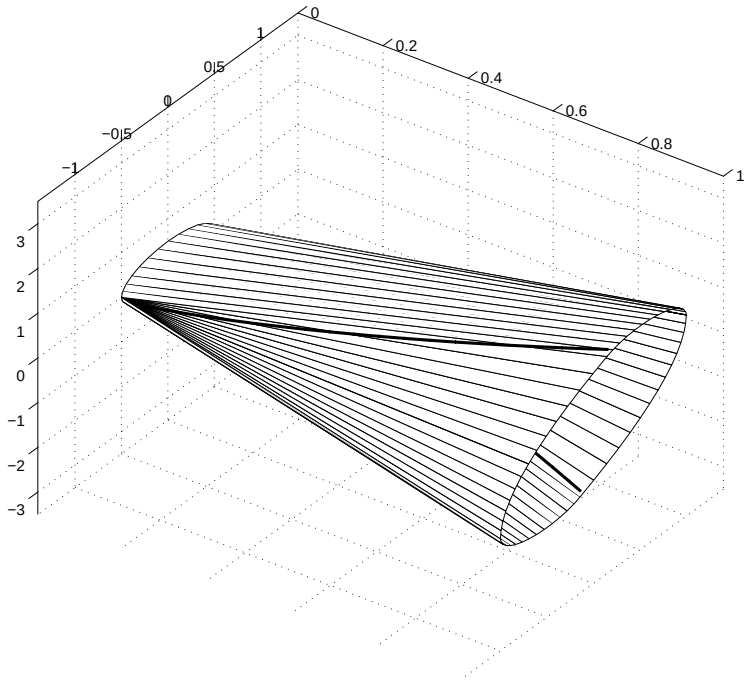


Figure 2

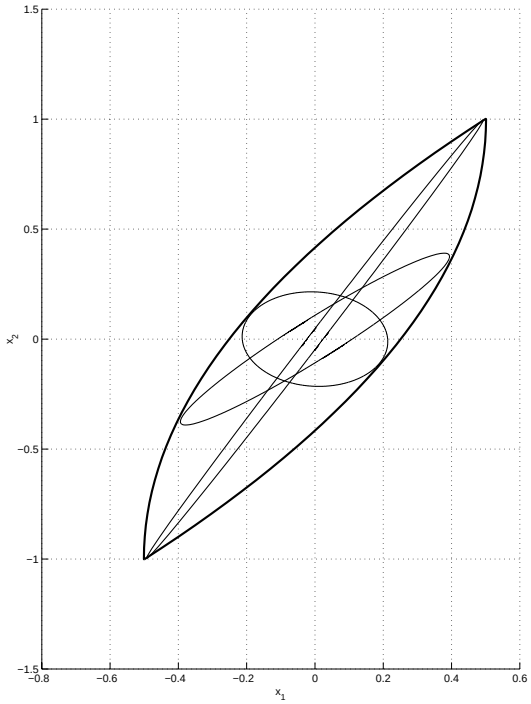


Figure 3

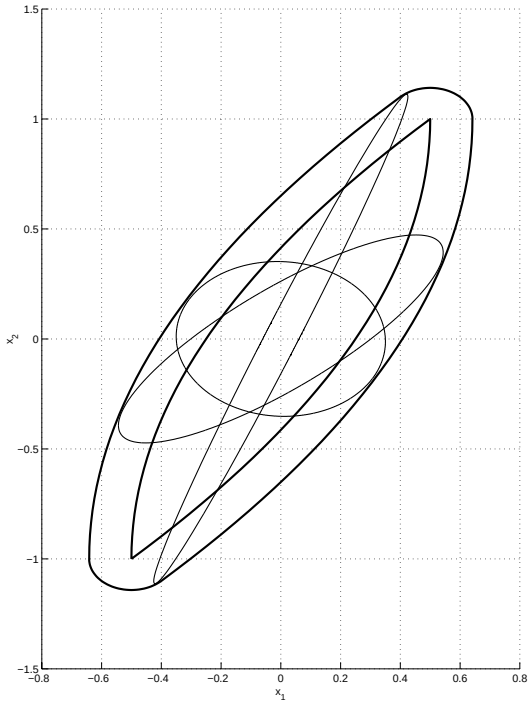
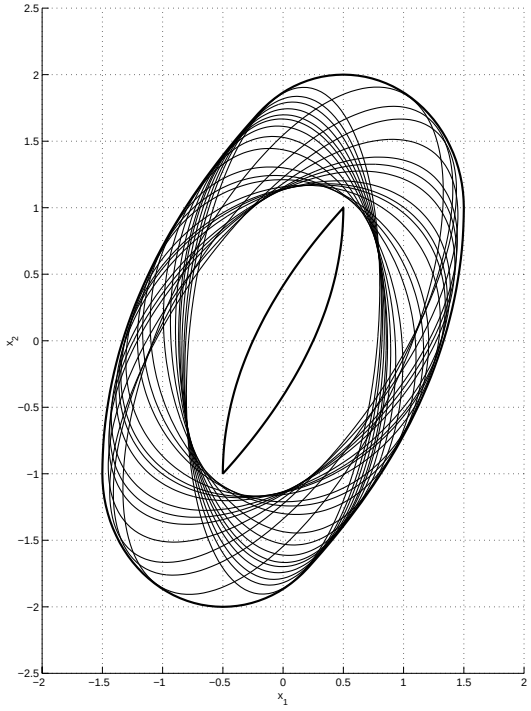


Figure 4



Conclusion

This paper studies tight internal ellipsoidal approximations of reach sets and reach tubes for time-varying linear systems. Equation (43) with appropriately chosen parameter $S(t)$ (an orthogonal matrix-valued function restricted by an equality) generates a family of internal ellipsoids that touch the reach tube *from inside* along a special family of "good" curves that cover the whole tube. It presents the relation for such "good" curves and develops techniques that allow on-line calculation of the internal ellipsoids without additional computational burden present in other approaches. The calculation of similar ellipsoids along any other given smooth curve on the boundary of the reach tube requires much more computation as compared with the "good" ones. The paper relies on relations different from those indicated in [1], [2], [8] and are relevant for solving various classes of control and design problems requiring guaranteed performance.

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